authorization

Service Description

Service ID: *"authorization"*

Abstract
This document describes the `authorization` service, version 1.0.0, produced by the ConsumerAuthorization core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `authorization` service, which provides provider-level authorization policies: a provider declares which consumers may use its service (or event type), and any system can ask whether a consumer is authorized. It is produced by the ConsumerAuthorization system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Successor of the Arrowhead 4.x Authorization `intracloud` rules. In 5.x a policy has a default rule and optional per-operation (scoped) rules, each `ALL`, `WHITELIST`, `BLACKLIST` or `SYS_METADATA`.

2.2 How This Service Is Meant to Be Used

A provider calls `grant` to publish the policy of one of its services and `revoke` to remove it; `lookup` lists its own policies. Orchestrators and providers call `verify` to evaluate a consumer request against the policies.

2.3 Important Delimitations and Dependencies

A provider can only set policies for itself (level `PROVIDER`). Management-level policies (level `MGMT`), set through `authorizationManagement`, take precedence: provider policies are only evaluated when no management policy applies. The service does not enforce anything; enforcement is done by the orchestrator (which only returns authorized providers) and by providers that check tokens.

3 Service Interface

This section describes the interfaces to the `authorization` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

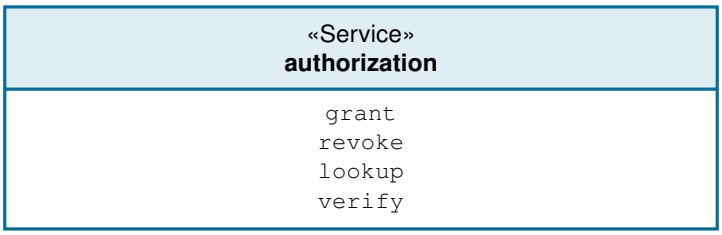


Figure 1: SysML-style block of the `authorization` service and its operations.

The following interface operations are available.

3.1 operation grant (AuthorizationGrantRequest) : AuthorizationPolicyResponse

Creates or replaces the provider-level policy of the requesting provider for one target. Returns 201 for a new policy, 200 if an identical one exists.

3.2 operation revoke (Identifier)

Removes a provider-level policy of the requester by instance identifier. Returns 200 if removed, 204 if it did not exist.

3.3 operation lookup (AuthorizationLookupRequest) : AuthorizationPolicyListResponse

Returns the requester's provider-level policies matching the filter.

3.4 operation verify (AuthorizationVerifyRequest) : Boolean

Evaluates whether the consumer may use the target (and operation) of the provider: management-level policies first, provider-level policies if none applies; no applicable policy means denied. Returns a boolean.

4 Information Model

Here, all data objects that can be part of the `authorization` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.9, which are used to represent things like hashes and identifiers.

4.1 struct AuthorizationGrantRequest

A provider-level authorization policy that the provider sets for its own service.

Field	Type	Description
cloud	Name	Cloud identifier of the consumer cloud the policy applies to; `LOCAL` for the own cloud.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Service definition (or event type) the policy protects.
description	String	Free-text description.
defaultPolicy	AuthorizationPolicyRequest	Policy that applies to all operations without a scoped policy.
scopedPolicies	Map<String, AuthorizationPolicyRequest>	Policies for individual operations, keyed by operation name.

4.2 struct AuthorizationLookupRequest

Filter for the requester's own provider-level policies.

Field	Type	Description
instanceIds	List<Identifier>	Policy instance identifiers to match.
cloudIdentifiers	List<String>	Cloud identifiers to match.
targetNames	List<String>	Target names to match.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.

4.3 struct AuthorizationPolicy

One policy rule.

Field	Type	Description
policyType	AuthorizationPolicyType	Policy type: `ALL`, `WHITELIST`, `BLACKLIST` or `SYS_METADATA`.
policyList	List<String>	System names of the white- or blacklist.
policyMetadata Requirement	MetadataRequirement	Metadata the consumer system must have (`SYS_METADATA`).

4.4 struct AuthorizationPolicyListResponse

List of authorization policies.

Field	Type	Description
entries	List<AuthorizationPolicyResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.5 struct AuthorizationPolicyRequest

One policy rule.

Field	Type	Description
policyType	String	Policy type: `ALL`, `WHITELIST`, `BLACKLIST` or `SYS_METADATA`.
policyList	List<String>	System names of the white- or blacklist.
policyMetadata Requirement	MetadataRequirement	Metadata the consumer system must have (`SYS_METADATA`).

4.6 struct AuthorizationPolicyResponse

A stored authorization policy.

Field	Type	Description
instanceId	Identifier	Policy instance identifier `<MGMT or PR>|<cloud>|<provider>|<targetType>|<target>`.
level	AuthorizationLevel	Policy level: `MGMT` (set by an operator) or `PROVIDER` (set by the provider itself).
cloud	Name	Cloud identifier of the consumer cloud the policy applies to; `LOCAL` for the own cloud.
provider	Name	Provider system the policy protects.
targetType	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
description	String	Free-text description.
defaultPolicy	AuthorizationPolicy	Policy that applies to all operations without a scoped policy.
scopedPolicies	Map<String, AuthorizationPolicy>	Policies for individual operations, keyed by operation name.
createdBy	Name	Name of the system that created the record.
createdAt	DateTime	UTC time of creation.

4.7 struct AuthorizationVerifyRequest

One authorization question: may the consumer use the target (operation) of the provider?

Field	Type	Description
provider	Name	Provider system; for `verify` defaults to the requester.
consumer	Name	Consumer system; for `verify` defaults to the requester.
cloud	Name	Cloud of the consumer, `LOCAL` by default.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
scope	Name	Operation to check; empty checks the default policy.

4.8 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

4.9 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
AuthorizationLevel	One of `MGMT`, `PROVIDER`.
AuthorizationPolicyType	One of `ALL`, `WHITELIST`, `BLACKLIST`, `SYS_METADATA`.
AuthorizationTargetType	One of `SERVICE_DEF`, `EVENT_TYPE`.
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “ConsumerAuthorization – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, consumer-authorization/documentation/v5.3.0/ConsumerAuthorization-v5.3.0-SysD.pdf.

[2] ——, “authorization HTTP/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorization-HTTP-IDD.pdf.

[3] ——, “authorization MQTT/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorization-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	